MALWANDLA HLONGWANE

Junior Software Developer | Full-Stack Web Developer

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PROFESSIONAL SUMMARY

Final-year Diploma in ICT (Applications Development) student at the University of Mpumalanga who builds and ships full-stack web applications. Projects include a live Node.js and MySQL marketplace with 134 automated tests, a Flutter and Supabase savings app live on Android and web, an enterprise Jakarta EE platform with CI/CD and Docker, and a security-hardened PHP MVC booking site. Skilled in REST API design, authentication and web security, relational database design, Git and automated testing. Seeking a Junior Software Developer, Graduate Developer or Software Development Internship role.

TECHNICAL SKILLS

- **Languages & Frameworks:** Java, JavaScript, PHP, SQL, HTML5, CSS3, Dart with Flutter (Riverpod, go_router)
- **Backend & APIs:** Node.js, Express.js, Jakarta EE (Faces, REST, EJB, JPA, JMS), Spring Boot, Supabase, REST API design, MVC and layered architecture
- **Databases:** MySQL, PostgreSQL, SQLite; schema design, normalisation, indexing, transactions, row-level security, versioned migrations
- **Security:** JWT, two-factor authentication (TOTP), bcrypt/PBKDF2 hashing, AES encryption, CSRF protection, Content-Security-Policy, rate limiting, input validation
- **Testing & DevOps:** Automated testing (Node.js test runner, JUnit 5, Flutter tests, pgTAP), Postman, Git, GitHub, GitHub Actions (CI), Docker, Maven, Render, Cloudflare Pages
- **Practices:** Agile/Scrum, SDLC, requirements (SRS), UML modelling, object-oriented design, design patterns
- **Professional:** Problem solving, teamwork on group projects, technical documentation, attention to detail, fast learner

PROJECT EXPERIENCE

Taskify | Campus Task & Equipment Marketplace

Full-stack web application | Deployed live
Tech: Node.js, Express, MySQL, JavaScript, HTML, CSS, REST API, JWT, Render

- Built and deployed a full-stack marketplace where students post tasks, hire services, buy items and rent equipment.
- Designed a 26-table MySQL schema (39 foreign keys, 76 indexes) behind a REST API with separate routes, controllers and services.
- Secured the platform with JWT authentication, TOTP two-factor sign-in for admins, bcrypt password hashing, Helmet security headers, rate limiting and server-side input validation.
- Stopped double bookings and duplicate payments under concurrent use with row-level locks and conditional updates.
- Wrote 134 automated tests across 13 test suites (Node.js test runner) covering rentals, sales, disputes and tasks, all passing.
- Integrated Cloudinary image uploads, Nodemailer email and web push notifications.

GitHub: github.com/Malwandla21-ict/taskify | **Live site:** taskify-marketplace.co.za

ZAKA | Savings Tracker & Spitori Dictionary

Cross-platform app for Android and web | Deployed live | Private repository, code available on request
Tech: Flutter, Dart, Riverpod, go_router, Supabase (PostgreSQL, Auth, Storage, Edge Functions), pgTAP, Cloudflare Pages

- Built and launched a savings app inspired by Pretoria's Spitori culture: savings goals with daily, weekly or monthly plans, ahead/behind tracking, payday reminders, charts, and a free Spitori dictionary.
- Designed a 17-table PostgreSQL schema in Supabase through 28 versioned migrations, with row-level security policies so each person can only read and change their own data.

- Wrote 435 Flutter tests and 198 pgTAP database checks covering row-level security, balance rules, limits and admin access.
- Shipped to production with a deploy script that builds the web app and signed Android APKs to Cloudflare Pages, plus security headers, custom account emails and POPIA privacy controls (download my data, delete my account).

Live site: zaka-3hi.pages.dev

StokVault | Stokvel Contribution & Payout Platform

Enterprise Java | Group project (DICT312) for UMP Centre for Entrepreneurship, Cooperatives & Innovation

Tech: Java 21, Jakarta EE 11 (Faces, REST, EJB, CDI, JPA, JMS, Batch), Payara Server, PostgreSQL, Maven, Docker, GitHub Actions, JUnit 5

- Delivered a layered Jakarta EE application that replaces WhatsApp and notebook record-keeping for stokvels, with role-based access for treasurers, committee members and members.
- Built a SHA-256 hash-chained, tamper-evident audit log, with a PostgreSQL trigger that blocks updates and deletes.
- Protected sensitive data with PBKDF2 password hashing, AES-256-GCM encryption of ID numbers and account lockout after failed logins.
- Set up GitHub Actions CI to build the app, run 49 JUnit 5 unit tests and 48 end-to-end API checks, and build a Docker image.

GitHub: github.com/Malwandla21-ict/StokVault

ADDITIONAL PROJECTS

UMP Events | Campus Event Booking Website

PHP web application | Started as a 2nd-year group project, since rebuilt

Tech: PHP 8.2, MySQL, PDO, MVC, HTML, CSS, JavaScript, Apache (XAMPP)

- Rebuilt a procedural PHP site into a framework-free MVC application with a router, middleware, controllers, services and repositories.
- Hardened security against common OWASP risks: prepared statements, CSRF tokens, a strict Content-Security-Policy, secure sessions, login throttling and safe file uploads.

GitHub: github.com/Malwandla21-ict/Ump-Event-Booking-web

FUEL SA | Fuel Ordering & Station Management System

Java desktop application

Tech: Java 17, JavaFX 21, FXML, CSS, SQLite (JDBC), Maven, BulkSMS REST API

- Built a desktop app where customers register vehicles and request fuel with a live cost estimate, and admins approve or reject requests from a dashboard.
- Integrated the BulkSMS REST API to send request status updates by SMS.

GitHub: github.com/Malwandla21-ict/FuelSA

STAMS | Student Accommodation Management System

Mobile app + REST API | In progress

Tech: Java 21, Spring Boot, Spring Security, JWT, Spring Data JPA, Hibernate, PostgreSQL, Flutter, Dart, Postman

- Building a Flutter mobile app and Spring Boot REST API for landlords to manage properties, rooms and student residents.

GitHub: github.com/Malwandla21-ict/STAMS-Student-Accomadation-Management-System-

EDUCATION

Diploma in Information and Communication Technology (Applications Development)

University of Mpumalanga | Expected completion: 2026 | Academic average: 75–78%

Relevant coursework:

- **Web Development:** PHP (OOP, forms, sessions, authentication), MySQL with PDO, JavaScript (DOM)
- **Application Development (Jakarta EE):** Servlets, JSF, RESTful web services, JSON binding, JPA/ORM, microservices
- **Software Engineering:** SDLC, requirements engineering, UML, architectural design, Agile/Scrum, software testing